Dissecting Mechanisms of Financial Crises: Intermediation and Sentiment

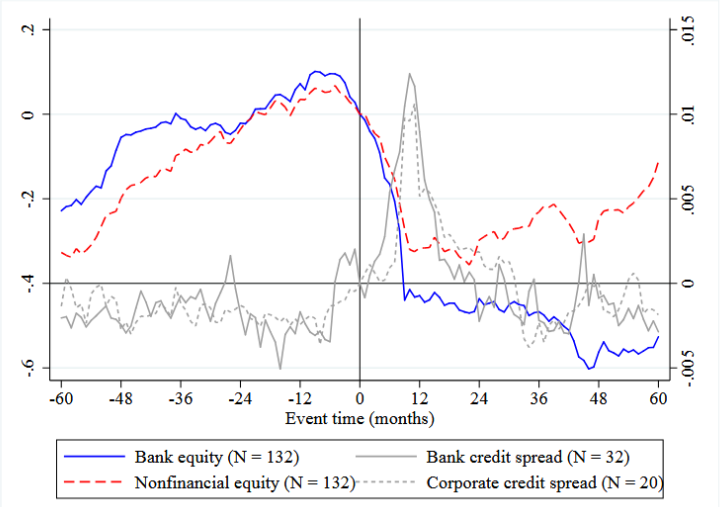
Krishnamurthy & Li *JPE* 2025

Félix Ranson

PSE macro reading group

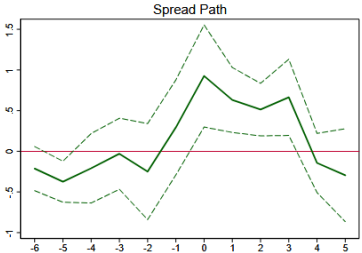
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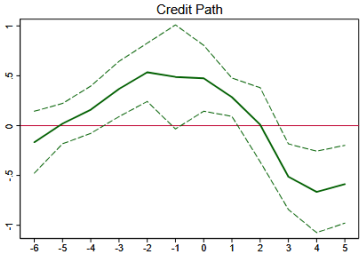
Motivation (1)

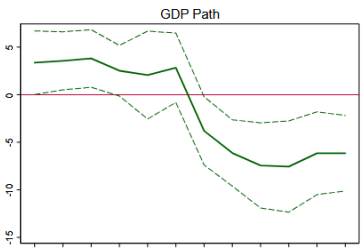


Taken from Khorrami & Mendo (2025), adapted from Baron et al. (2021)

Motivation (1)







Taken from Khrisnamurthy & Muir (2025)

Motivation (2)

Financial accelerator:

- BBG 1999: disruptions in financial intermediation increase spreads and amplify shocks to real activity
- BS 2014: Non-linear amplification of asset prices means that the economy is prone to occasional volatility crisis episodes + volatility paradox
- Cylical properties of these models are at odds with the patterns above

$\hookrightarrow$ On its own, the financial accelerator cannot explain the endogenous build-up of fragilities

This paper: Adding belief formation helps match the patterns around financial crises

Overview of results:

- Frictional intermediation is necessary to matches crisis dynamics
- Belief formation generates high credit prior to crisis and tightened spreads (froth)

Model ingredients: agents and production

- Environment: continuous-time AK economy, as in Brunnermeier–Sannikov (2014)
- Agents: a continuum of bankers and households with discounted log utility and nonnegative wealth
- Intermediation: bankers operate capital more productively than households, $\bar{A} > \underline{A}$
- Capital accumulation: investment, depreciation at rate δ, and aggregate Brownian shocks $\sigma^K dB_t$

$$\frac{dk_{j,t}}{k_{j,t}} = \mu_t^K dt - \delta dt + \sigma^K dB_t$$

- Investment: convex adjustment costs $\phi(\mu^K)$: the capital price p_t determines the investment rate $p_t = \phi'(\mu_t^K)$
- Banker turnover: bankers become households at rate η, transferring their wealth out of the banking sector

Model ingredients: intermediation and beliefs

- Financing: banks use their own net worth and runnable short-term debt, no outside equity or long-term debt
- Liquidity shocks: Poisson events dN_t trigger funding withdrawals and a liquidation cost α, paid to households
- Amplification: fire-sale price declines erode bank net worth and shift capital to less productive households
- Hidden risk: the shock intensity switches between λ_L and λ_H: agents observe run events, not the regime
- Beliefs: common across agents: static, Bayesian, or diagnostic expectations about liquidity risk

Beliefs: learning about a hidden regime

The illiquidity shock dN_t arrives at an unobserved intensity $\widetilde{\lambda}_t \in \{\lambda_L, \lambda_H\}$ with $\lambda_L < \lambda_H$

The regime itself changes:

$$\lambda_L \xrightarrow{\lambda_{L\to H}} \lambda_H \qquad \lambda_H \xrightarrow{\lambda_{H\to L}} \lambda_L$$

Bankers and households observe the same history $\mathcal{F}_t^N = \sigma(N_s : s \leq t)$ and form the same beliefs

Define

$$\pi_t = \Pr(\widetilde{\lambda}_t = \lambda_H \mid \mathcal{F}_t^N) \qquad \lambda_t = \mathbb{E}[\widetilde{\lambda}_t \mid \mathcal{F}_t^N] = \pi_t \lambda_H + (1 - \pi_t)\lambda_L$$

- A long period without a run is evidence for the low-intensity regime
- A run is evidence for the high-intensity regime

$\hookrightarrow$ Beliefs concern the likelihood of another liquidity shock. The occurence of a run need not mean that the hidden regime has just changed

Bayesian updating: bad news and no news

If a run occurs, Bayes's rule gives

$$\pi_t = \frac{\lambda_H \pi_{t^-}}{\lambda_{t^-}} \quad \Longrightarrow \quad \Delta\lambda_t = \kappa^\lambda(\lambda_{t^-}) = \frac{(\lambda_{t^-} - \lambda_L)(\lambda_H - \lambda_{t^-})}{\lambda_{t^-}} > 0$$

Between runs, both regime switching and the absence of a run change beliefs:

$$d\lambda_t = \mu^\lambda(\lambda_{t^-})\, dt + \kappa^\lambda(\lambda_{t^-})\, dN_t$$

$$\mu^\lambda(\lambda) = \underbrace{(\lambda_L - \lambda)\lambda_{H\to L} + (\lambda_H - \lambda)\lambda_{L\to H}}_{\text{hidden regime switches}} - \underbrace{(\lambda - \lambda_L)(\lambda_H - \lambda)}_{\text{no run is good news}}$$

$\hookrightarrow$ After a run, perceived risk gradually recedes during quiet times ($\mu^\lambda > 0$ is transient)

Diagnostic expectations

What if agents overweight states that have become more likely? Compare the Bayesian posterior π_t with the forecast made T years ago:

$$\pi_t^T = \mathbb{E}[\pi_t \mid \mathcal{F}_{t-T}^N] \qquad \pi_t^\theta = \frac{\pi_t(\pi_t/\pi_t^T)^\theta}{Z_t} \qquad 1-\pi_t^\theta = \frac{(1-\pi_t)[(1-\pi_t)/(1-\pi_t^T)]^\theta}{Z_t}$$

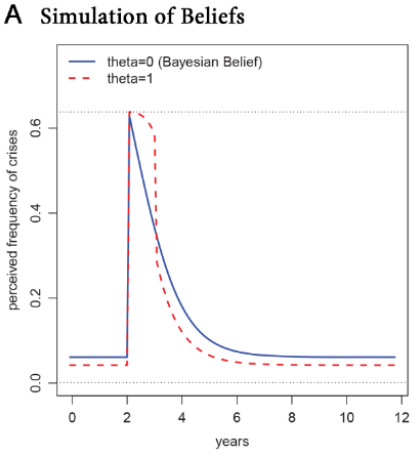
Here Z_t makes the two probabilities sum to one. Equivalently,

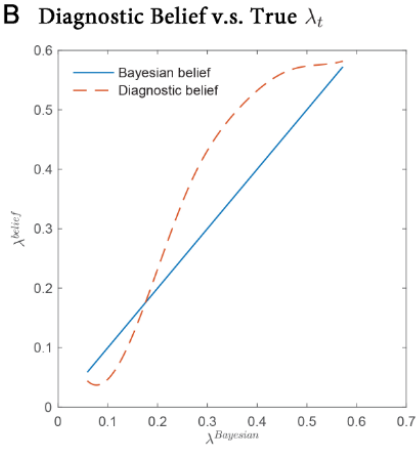
$$\frac{\pi_t^\theta}{1-\pi_t^\theta} = \frac{\pi_t}{1-\pi_t}\left[\frac{\pi_t/(1-\pi_t)}{\pi_t^T/(1-\pi_t^T)}\right]^\theta \qquad \lambda_t^\theta = \lambda_L + (\lambda_H - \lambda_L)\pi_t^\theta$$

- $\theta = 0$: Bayesian beliefs
- $\theta > 0$: amplify the change in odds relative to the reference forecast

$\hookrightarrow$ The reference is a forecast for today, not simply the old posterior. The calibration uses $\theta = 0.9$ and $T = 1$ year

Overoptimism, then overpessimism





The illustration uses $\theta = 1$. Quiet periods lower perceived risk, a run raises it sharply. Diagnostic beliefs can exaggerate both movements

Intermediaton: banker's problem

Throughout we will denote by x_{t-} a coefficient that is fixed before a jump occurs

$$V^b(w_t^b, w_t, \lambda_t) = \sup_{c_t^b \geq 0,\, x_{t-}^K \geq 0} E^{\mathbb{Q}} \left[\int_t^\tau e^{-\rho(s-t)} \log(c_s^b)\, ds + e^{-\rho\tau} V^h(w_\tau^b, w_\tau) \,\middle|\, w_t^b, w_t \right]$$

s.t. $w_t^b \geq 0$

$$\frac{dw_t^b}{w_{t-}^b} = \underbrace{x_{t-}^K \left(\mu_{t-}^R + \frac{\bar{A}}{p_{t-}} \right) dt}_{\text{exp. return from capital}} - \underbrace{x_{t-}^d r_{t-}^d\, dt}_{\text{deposit funding}} - \underbrace{\frac{c_{t-}^b}{w_{t-}^b} dt}_{\text{consumption}} + \underbrace{x_{t-}^K \left(\sigma^K + \sigma_{t-}^p \right) dB_t}_{\text{volatility of capital}}$$

$$- \underbrace{\left(x_{t-}^K \kappa_{t-}^p + \alpha \left(x_{t-}^d \right)^+ \right)}_{k^b = \text{losses in illiquidity shock}} dN_t$$

and where $\tau \sim \mathcal{E}(\eta)$

Intermediaton: HH's problem

$$V^h(w_t^h, w_t, \lambda_t) = \sup_{c_t^h \geq 0,\, y_{t-}^K \geq 0} E^{\mathbb{Q}} \left[\int_t^\infty e^{-\rho(s-t)} \ln(c_s^h)\, ds \,\middle|\, w_t^h, w_t \right]$$

$$\text{s.t. } w_t^h \geq 0$$

$$\frac{dw_t^h}{w_{t-}^h} = \underbrace{y_{t-}^K \left(\mu_{t-}^R + \frac{A}{p_{t-}} \right) dt}_{\text{exp. return from capital}} + \underbrace{y_{t-}^d r_{t-}^d\, dt}_{\text{deposit interest}} - \underbrace{\frac{c_t^h}{w_{t-}^h} dt}_{\text{consumption}}$$

$$+ \underbrace{y_{t-}^K \left(\sigma^K + \sigma_{t-}^p \right) dB_t}_{\text{volatility of capital}} - \underbrace{\kappa_{t-}^h\, dN_t}_{\text{liquidity exposure}}$$

$$\kappa_{t-}^h = \underbrace{y_{t-}^K \kappa_{t-}^p}_{\text{valuation drop}} \underbrace{- \alpha \left(x_{t-}^d \right)^+ \frac{w_{t-}}{1 - w_{t-}}}_{\text{transfer}}$$

Amplification: characterizing the equilibrium

Define $w_t = \frac{w_t^b}{w_t^h + w_t^b}$ the banker's wealth share and $\psi_t = \frac{x_t^K w_t^b}{x_t^K w_t^b + y_t^K w_t^h}$ the share of capital that they operate

Definition

A fundamental equilibrium that is Markov in w_t, λ_t is a set of functions $p(w_t, \lambda_t)$, $r^d(w_t, \lambda_t)$, $\hat{c}^h(w_t, \lambda_t)$, $y^K(w_t, \lambda_t)$, $\hat{c}^b(w_t, \lambda_t)$, $x^k(w_t, \lambda_t)$ s.t.:

1. consumption, investment and portfolio choices are optimal
2. $W_t^b x_t^k + W_t y_t^k = p_t K_t$
3. $W_t^b + W_t^h = p_t K_t$
4. $\hat{c}^h W_t^h + \hat{c}^b W_t^b = (\psi_t \bar{A} + (1 - \psi_t)\underline{A})K_t - (\mu_t^k + \phi(\mu_t^k))K_t$

Equilibrium characterization

Intermediaries always take on leverage such that $x_t^k \geq 1$ and $(x_{t^-}^d)^+ = x_{t^-}^d$. We can rewrite the budget constraint

$$\frac{dw^b}{w_-^b} = \left[-\frac{c^b}{w_-^b} + r^d + x^K \left(\mu^R + \frac{\bar{A}}{p} - r^d \right) \right] dt + x^K (\sigma^K + \sigma^p)\, dB_t - \left(\alpha x^d + x^K \kappa^p \right) dN_t, \qquad x^d = x^K - 1.$$

With the usual results on log utility[1] we get the objective (partial HJB):

$$\sup_{\substack{c^b>0,\ x^K \geq 1 \\ x^d = x^K - 1}} \left\{ \log c^b + \frac{1}{\rho} \left[-\frac{c^b}{w^b} + r^d + x^K \left(\mu^R + \frac{\bar{A}}{p} - r^d \right) - \frac{1}{2}(x^K)^2 (\sigma^K + \sigma^p)^2 + \lambda \log\left(1 - \alpha x^d - x^K \kappa^p\right) \right] \right\}$$

$\hookrightarrow$ Mean-Variance objective + a jump illiquidity term

[1] $c = \rho w^b$ and $V^b(w, z) = \rho^{-1} \log w + v^b(z)$

Equilibrium characterization

Portfolio choice of bankers characterized by the martingale condition

$$\mathbb{E}^{\mathbb{Q}}_{t^-}[dR^b_t] - r^d_{t^-} - \lambda_{t^-}(\alpha x^d_{t^-} + x^K_{t^-}\kappa^p_{t^-}) = \underbrace{\varsigma^b_{t^-}}_{\substack{\text{price of}\\ \text{Brownian risk}}} (\sigma^K + \sigma^p_{t^-}) + \lambda \underbrace{\nu^b_{t^-}}_{\substack{\text{price of}\\ \text{liquidity risk}}} (\alpha x^d_{t^-} + x^K_{t^-}\kappa^p_{t^-})$$

$$= \underbrace{(\sigma^K + \sigma^p_{t^-})^2 x^k_{t^-}}_{\text{Brownian risk premium}} + \underbrace{\lambda_{t^-}(\alpha + \kappa^p_{t^-}) \frac{\alpha x^d_{t^-} + x^K_{t^-}\kappa^p_{t^-}}{1 - \alpha x^d_{t^-} - x^K_{t^-}\kappa^p_{t^-}}}_{\text{liquidity risk premium}}$$

For a household we get

$$\mathbb{E}^{\mathbb{Q}}_{t^-}[dR^b_t] - r^d \leq (\sigma^K + \sigma^p_{t^-})^2 y^K + \lambda \frac{\kappa^p}{1 - \kappa^h} \qquad \text{with equality if } y^k > 0$$

Solution method: step (1)

Express the unknowns in terms of the price function $p(w, \lambda)$

$$\rho = \frac{\psi \bar{A} + (1-\psi)\underline{A} - (\mu^K + \phi(\mu^K))}{p} \qquad \text{(PO)}$$
$$x^K w + y^k (1-w) = 1$$
$$\psi = x^K w$$

$\hookrightarrow$ For a given state (w, p) and p at this state, we know x^K, y^K and $i = \mu^K(p) + \phi(\mu^K(p))$

Substracting the portfolio choice conditions gives:

$$\frac{\bar{A} - \underline{A}}{p} \geq (\sigma)^2 (x^K - y^K) + \lambda \left[\frac{\kappa^p + \alpha}{1-\kappa^b} - \frac{\kappa^p}{1-\kappa^h} \right] \quad \text{with equality if } y^K > 0 \qquad \text{(RB)}$$

There are two regimes: a "crisis" regime when $y^K > 0$ and (RB) holds and a "normal" regime where $y^K = 0$, $\psi = 1$ and $\bar{p}$ satisfies $\rho \bar{p} = \bar{A} - i(\bar{p})$

Solution method: step (2)

Derive the law of motions of the states:

$$\begin{aligned}\frac{dw}{w} &\triangleq \mu^w\,dt + \sigma^w\,dB - \kappa^w\,dN \\ &= (1-w)\left(\mu^b - \mu^h + (\sigma^h)^2 - \sigma^b\sigma^h - w(\sigma^b-\sigma^h)^2 - \eta\right)dt \\ &\quad + (1-w)(\sigma^b-\sigma^h)\,dB - (1-w)\frac{1-\frac{1-\kappa^b}{1-\kappa^h}}{1+w\left(\frac{1-\kappa^b}{1-\kappa^h}-1\right)}\,dN\end{aligned}$$

With $\mu^j, \sigma^j, \kappa^j$ given by the type$-j$ wealth process

Recall that of the Bayesian belief process:

$$d\lambda_t = \mu^\lambda(\lambda_{t^-})\,dt + \kappa^\lambda(\lambda_{t^-})\,dN_t$$

The same run simultaneously changes bank wealth and beliefs

Solution method: step (3)

Assume $p = p(w, \lambda)$ and apply Ito's lemma$\rightarrow$a system of ODEs and fixed-point conditions

$$\mu^p = \frac{1}{p}\left[p_w\, w\mu^w + \frac{1}{2}p_{ww}(w\sigma^w)^2 + p_\lambda \mu^\lambda\right]$$

$$\sigma^p = \frac{p_w}{p}\, w\sigma^w = \frac{p_w}{p}\, w(1-w)(\sigma^b - \sigma^h)$$

$$\kappa^p = 1 - \frac{p\big(w(1-\kappa^w),\, \lambda + \kappa^\lambda\big)}{p(w, \lambda)}$$

$$\kappa^b = x^K \kappa^p + \alpha x^d$$

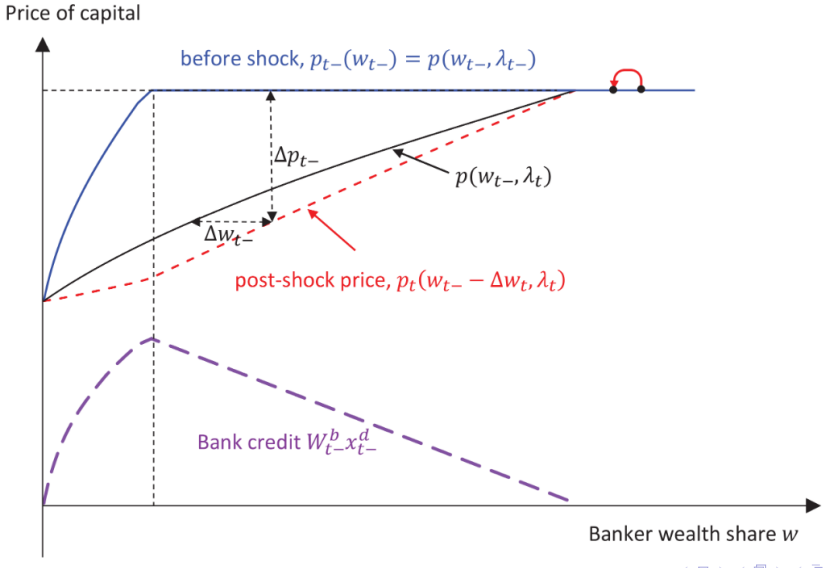
$$\kappa^h = y^K \kappa^p - \alpha x^d \frac{w}{1-w}$$

$$\kappa^w = \frac{(1-w)(\kappa^b - \kappa^h)}{1 - \kappa^h - w(\kappa^b - \kappa^h)}$$

$$\kappa^\lambda = \frac{(\lambda - \lambda_L)(\lambda_H - \lambda)}{\lambda}$$

Just solve! Set $p = \bar{p}$ when $\psi = 1$

State-dependent amplification



Taking the model to the data

1. Bank credit: runnable debt issued to households, relative to output

$$\frac{\text{Credit}_t}{Y_t} = \frac{W_t^b x_t^d}{Y_t} = \frac{p_t w_t x_t^d}{\psi_t \bar{A} + (1-\psi_t)\underline{A}}$$

2. Liquidity spread: a hypothetical instantaneous loan not exposed to dB_t, but with loss α in a run

$$L_t = r_t^C - r_t^d = \frac{\alpha \lambda_t}{1-\kappa_t^b} \qquad \kappa_t^b = x_t^K \kappa_t^p + \alpha x_t^d$$

3. Credit spread: price zero-net-supply risky and safe bonds with the banker's pricing kernel and the same expected maturity τ

Need a notion of default to map to the BAA bond spread

4. Financial crisis: Not just an illiquidity jump! Monthly growth in bank credit/GDP in the bottom 4% of its simulated distribution, with no other such event in the preceding three years.

Calibration

Externally set: $\delta = 0.10$ $\rho = 0.04$ $\chi = 3$ $\alpha = 0.05$ $\theta = 0.90$ $\lambda_L = 0.001$ $T = 1$

Targeted moments

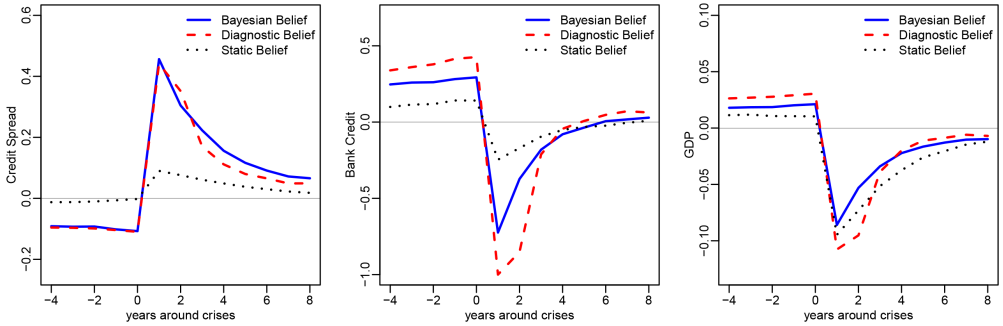
	Data	Static	Bayes	Diag.
Liquidity spread (%)	.90	.86	.77	.70
Crisis spread rise (%)	70	9	56	55
Spread half-life (years)	2.5	3.4	2.5	2.5
Output/capital (%)	14	16	17	15
3-year crisis GDP (%)	−9.1	−8.9	−7.9	−9.2
GDP growth vol. (%)	3.8	3.6	2.9	3.8
Bank leverage	5.0	5.1	5.1	5.0

Internally chosen parameters

	Static	Bayes	Diag.
$\bar{\lambda}$	.072	—	—
λ_H	—	.561	.638
$\lambda_{L\to H}$	—	.11	.11
$\lambda_{H\to L}$	—	.47	.48
$\underline{A}$	.12	.17	.13
$\bar{A} - \underline{A}$	.055	.030	.024
σ^K	.06	.03	.03
η	.122	.055	.034

$\hookrightarrow$ Recalibrate each version to the same targets. Liquidity spreads disciplines $\tilde{\lambda}_t$, crisis output losses discipline the productivity gap ($\bar{A} - \underline{A}$), average leverage disciplines banker exit η

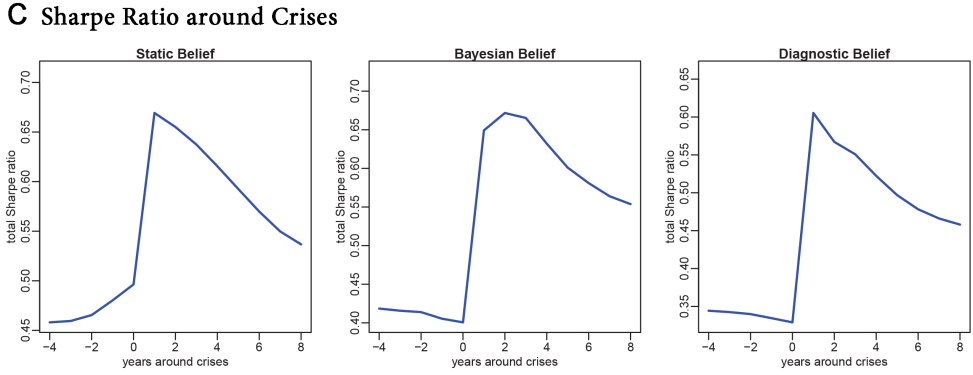
Does the model reproduce the crisis cycle?



- Crisis and recovery: spreads jump, credit contracts and GDP falls, recovery takes time in all three models
- Before the crisis: rising credit and low spreads appear together only with changing beliefs

(Quantities still overshoot: the Bayesian credit contraction is about $0.6\sigma//0.33\sigma$ in the data)

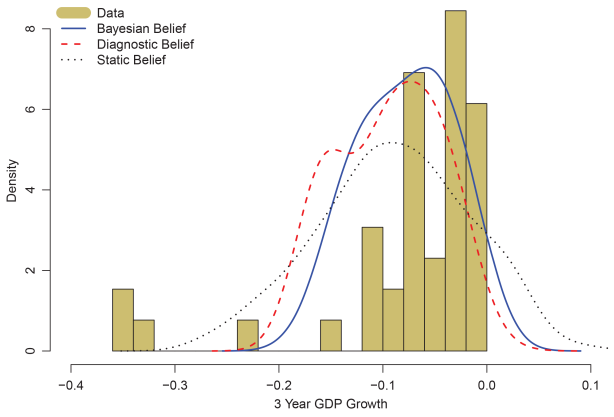
Risk premia: the calm before the storm?



All three models generate higher risk premia and volatility during crises. With dynamic beliefs, both also decline during the preceding boom. With static beliefs, they rise as fragility builds.

↪ Low perceived risk can coincide with rising exposure to a tail event: countercyclical volatility

Crisis and post crisis: the role of frictional intermediation



Even the static-belief model generates a distribution with a large left tail

The same liquidity shock has very different effects depending on banks' exposure and the strength of fire-sale amplification

$\hookrightarrow$ Changing beliefs are not needed to obtain severe crisis dynamics

Model: three-year GDP growth after crises. Data: peak-to-trough GDP growth. These are related, but not identical, measures

What can intermediation alone explain?

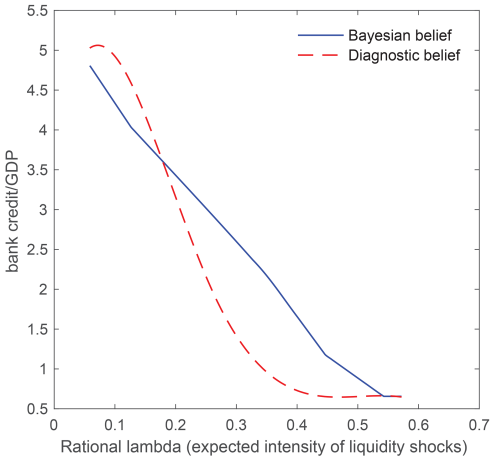
Untargeted regressions: compare credit conditions with subsequent outcomes

	Static	Bayes	Diag.	Data
Three-year GDP growth during crises				
Spread increase $\times$ crisis	-4.88	-2.87	-3.44	-2.11
Bank credit/GDP $\times$ crisis	-0.98	-2.18	-3.49	-2.06
Next-year excess returns				
Bank credit/GDP	-0.01	-0.01	-0.02	-0.02

A banking sector with little risk-bearing capacity lends less and requires higher risk premia. Large credit exposures also amplify losses in a crisis.

$\hookrightarrow$ These signs appear even with static beliefs. The credit–risk-premium relation alone does not identify a sentiment mechanism

Beliefs: low perceived risk encourages borrowing



Hold bank wealth share w fixed. Lower perceived run risk makes debt-funded capital more attractive:

$$\lambda \downarrow \implies \begin{cases} \text{bank credit} \uparrow \\ \text{spreads} \downarrow \end{cases}$$

More credit means greater losses if a run occurs.

Diagnostic beliefs steepen this relationship: optimism at low λ, pessimism at high λ

$\hookrightarrow$ The horizontal axis is the Bayesian filtered intensity//the unobserved regime $\widetilde{\lambda}$

Explaining pre-recrisis froth

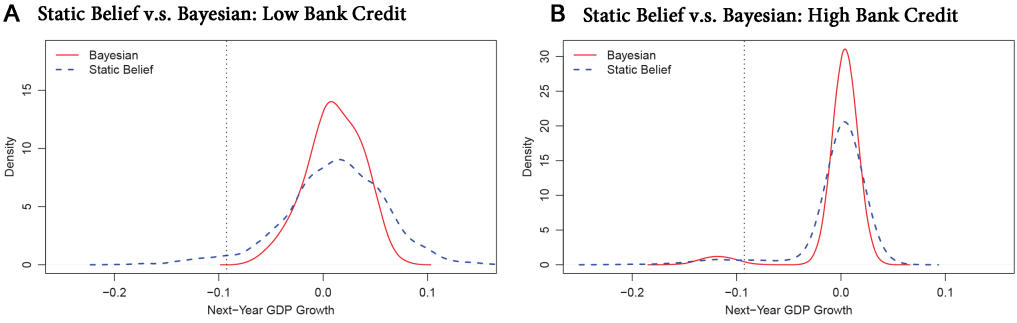
Can low spreads precede crises and help predict them?

	Static	Bayes	Diag.	Data
Precrisis spread (σ_S)	$+0.25$	-0.25	-0.30	-0.44
Froth $\rightarrow$ crisis, next 3 years (pp)	-5.94	$+5.67$	$+7.40$	$+12.90$
Credit $\rightarrow$ crisis, next year (pp)	$+0.13$	$+4.05$	$+3.85$	$+2.11$

With static beliefs, forward-looking spreads reflect the rising severity of a possible crisis. They are high, rather than low, before crises.

$\hookrightarrow$ Belief variation gets the froth relation's sign right. Diagnostic expectations move its magnitude closer to the data, but neither version fully matches it.

Less frequent runs, but more financial crises?



Two forces compete: lower λ means fewer runs, but also more borrowing and greater amplification when a run occurs.

In the calibrated belief models, the second force dominates High-credit states have a tighter central distribution of GDP growth and more mass in the crisis tail

Explaining tightening spreads

Recall that a robust feature of the data is that tight spreads predict crises

The bond used to measure spreads has default losses

$$\mathrm{LGD}_t = \widehat{\kappa}_0 + m\kappa_t^p$$

There are two opposing effects on its spread:

- Lower expected shock frequency $\longrightarrow$ fewer expected defaults $\longrightarrow$ lower spreads
- Greater fragility $\longrightarrow$ larger capital-price falls in a crisis $\longrightarrow$ higher losses given default $\longrightarrow$ higher spreads

The second effect weakens the froth relation. Its sign and magnitude depend on the cyclicality of recoveries, calibrated from historical BAA defaults.

$\hookrightarrow$ High credit predicting crises does not automatically imply that low spreads predict crises. The mapping from the model to the bond data matters for that result

Could beliefs do the job without financial frictions?

In the appendix, allow banks to issue equity and share risk with households. To give the jump a real effect, replace the liquidity transfer with direct destruction of productive capital:

$$\frac{dK_t}{K_{t^-}} = (\mu_t^K - \delta)\, dt + \sigma^K\, dB_t - \kappa_t^K\, dN_t$$

With identical log preferences and no financial constraints, all capital is operated at productivity $\bar{A}$. Goods-market clearing gives

$$\rho p_t = \bar{A} - i(p_t) \quad \Longrightarrow \quad p_t = \bar{p}$$

Changes in risk premia are offset by changes in the risk-free rate. Bank credit/GDP is constant in this frictionless benchmark.

$\hookrightarrow$ Beliefs about disasters alone cannot reproduce the joint movements of credit, asset values and crisis severity generated by impaired intermediation

Comment (1): "Sentiment" vs coordination

The Markovian restriction does a lot of work:

- We have more unknowns than equations, hence additional degrees of freedom in constructing the equilibrium
- Imposing $p = p(w, .)$ and applying Ito's lemma eliminates these degrees of freedom
- If one does not impose this restriction, $\sigma^p, \mu^p, \kappa^p$ can be determined by expectation coordination (Khorrami-Mendo 2025)

$\hookrightarrow$ Even without the belief process, the model already features indeterminacy that can be leveraged to give rise to the crisis patterns: self-fulfilling expectations about endogenous risk//Learning about exogenous risk

- A much more interesting approach would be to use equilibrium refinements techniques (adaptive learning, global games) to characterize better the set of credible equilibria and see if they give rise to the required patterns

Comment (2): which financial frictions?

The need for belief variations to match crisis moments hinges on the "strange" cyclical properties of the financial accelerator

- Countercyclical leverage
- The bigger the amplification, the faster the recovery because spreads help recapitalize bankers

Could also add other intermediation states:

- Risk-budgeting makes intermediary leverage a state and yields a very different picture (Adrian & Shin 2014, Adrian & Boyarchenko 2015, Coimba & Rey 2024)
- Slow-moving intermediation capacity (Gopalakrishna 2026)

Bottom line: Yes there are missing states, not sure at all that they are related to belief formation nor that other intermediation frictions could not generate froth too

La Fin